

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING INTERNSHIP

TASK 6

Deep Learning Fundamentals & Neural Network Implementation

Project: Breast Cancer Classification using an Artificial Neural Network

Internship Organization	Maincrafts
Project Type	Deep Learning / Binary Classification
Framework	TensorFlow / Keras
Dataset	Scikit-learn Breast Cancer Dataset

Prepared for Task 6 internship submission

1. Executive Summary

This project presents an end-to-end Deep Learning workflow for binary classification using the Breast Cancer dataset from Scikit-learn. An Artificial Neural Network (ANN) was developed with TensorFlow/Keras and evaluated using test accuracy, loss, precision, recall, F1-score and a confusion matrix. Training and validation curves were also analyzed.

The ANN achieved a test accuracy of **94.74%** with a test loss of **0.105609**. For comparison, Logistic Regression achieved **98.25%** accuracy on the same recorded evaluation split. Therefore, Logistic Regression performed better for this particular experiment, while the ANN successfully demonstrated the required deep-learning implementation.

2. Introduction

Deep learning is a branch of machine learning that uses neural networks with multiple layers to learn patterns and representations from data. Artificial Neural Networks consist of interconnected neurons organized into input, hidden and output layers. Model parameters are learned during training by minimizing a loss function.

The Task 6 project applies these concepts to a real classification problem. The workflow includes data preprocessing, feature scaling, ANN design, activation functions, training, regularization, evaluation, visualization and comparison with a traditional machine-learning baseline.

3. Objectives

- Understand deep learning and Artificial Neural Network fundamentals.
- Understand forward propagation, backpropagation and gradient-based optimization.
- Use ReLU and Sigmoid activation functions.
- Implement and train an ANN using TensorFlow/Keras.
- Preprocess and scale numerical features before training.
- Evaluate classification performance using multiple metrics.
- Visualize training and validation performance.
- Use Dropout as a regularization technique.
- Compare ANN performance with Logistic Regression.

4. Dataset

The Breast Cancer dataset from Scikit-learn was used. It contains **569 samples** and **30 numerical features** for a binary classification problem.

Property	Value
Dataset	Scikit-learn Breast Cancer Dataset
Samples	569
Features	30
Target shape	(569,)
Problem type	Binary classification
Train-test split	80% / 20%
Feature scaling	StandardScaler

5. Technologies and Tools

Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, Jupyter Notebook, VS Code, Git and GitHub.

6. Project Workflow

- Load the Breast Cancer dataset.
- Separate input features and target labels.
- Split the data into training and test sets.
- Apply StandardScaler to numerical features.
- Build the ANN using Dense and Dropout layers.
- Compile the model with Adam and Binary Crossentropy.
- Train for 50 epochs using batch size 32 and 20% validation split.
- Evaluate the trained model on the test set.
- Generate classification metrics and a confusion matrix.
- Plot training/validation accuracy and loss.
- Compare ANN with Logistic Regression.
- Save the trained model and scaler.

7. Data Preprocessing

The dataset was divided into training and testing subsets using an 80:20 split with stratification. StandardScaler was fitted on the training data and used to transform both training and test features. This prevents the feature scales from differing substantially and supports stable neural-network optimization.

8. Artificial Neural Network Architecture

Layer	Configuration	Purpose
Input	30 features	Receives scaled features
Dense 1	32 neurons, ReLU	Learns nonlinear representations
Dense 2	16 neurons, ReLU	Learns higher-level representations
Dropout	20%	Regularization
Output	1 neuron, Sigmoid	Binary classification output

Architecture flow: Input (30) → Dense (32, ReLU) → Dense (16, ReLU) → Dropout (20%) → Output (1, Sigmoid)

9. Activation Functions

ReLU: ReLU (Rectified Linear Unit) was used in the hidden layers to introduce non-linearity. **Sigmoid:** Sigmoid was used in the output layer because the task is binary classification.

10. Forward Propagation, Backpropagation and Gradient Descent

Forward propagation passes input features through the network to produce predictions. **Backpropagation** calculates how the loss changes with respect to model parameters and propagates this information backward. The optimizer then updates parameters to reduce the loss. Adam was used in this project.

11. Model Training Configuration

Parameter	Value
Framework	TensorFlow / Keras
Optimizer	Adam
Loss function	Binary Crossentropy
Metric	Accuracy
Epochs	50
Batch size	32
Validation split	20%
Dropout	20%

12. Training and Validation Performance

The training curves show rapid improvement during the early epochs. Training and validation accuracy increase toward approximately 0.99–1.00, while training and validation loss decrease substantially and remain close to each other near the end of training. This indicates strong learning performance on the recorded training/validation process without a large visible separation between the two curves.

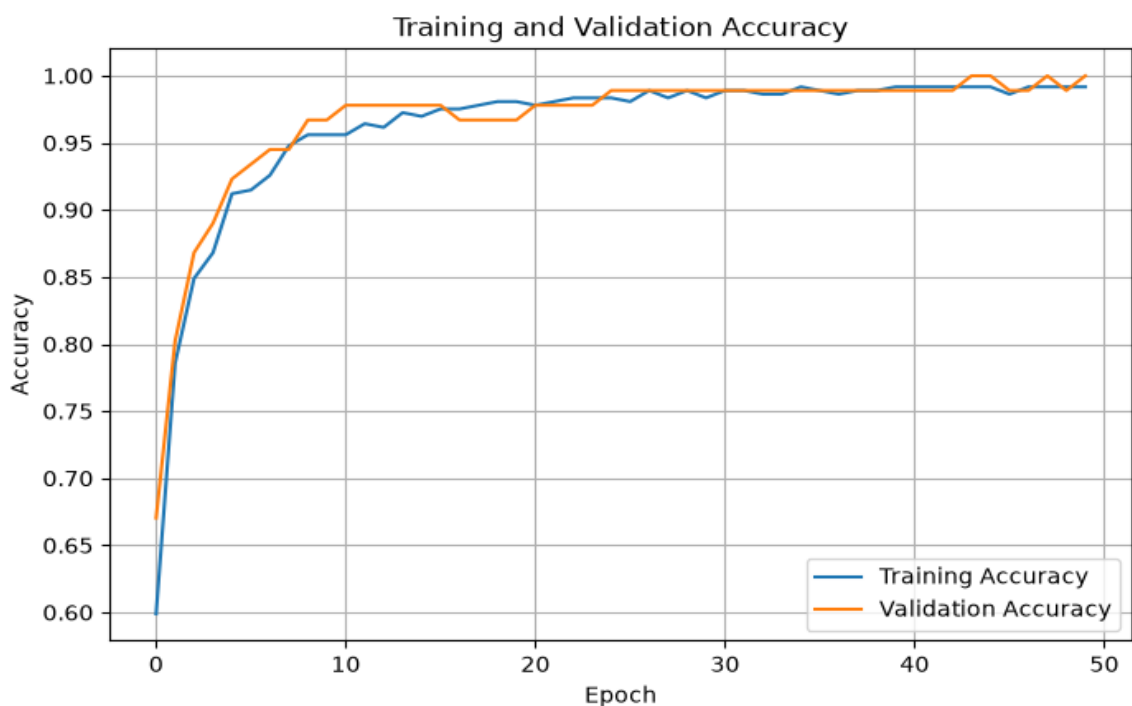


Figure 1. Training and Validation Accuracy

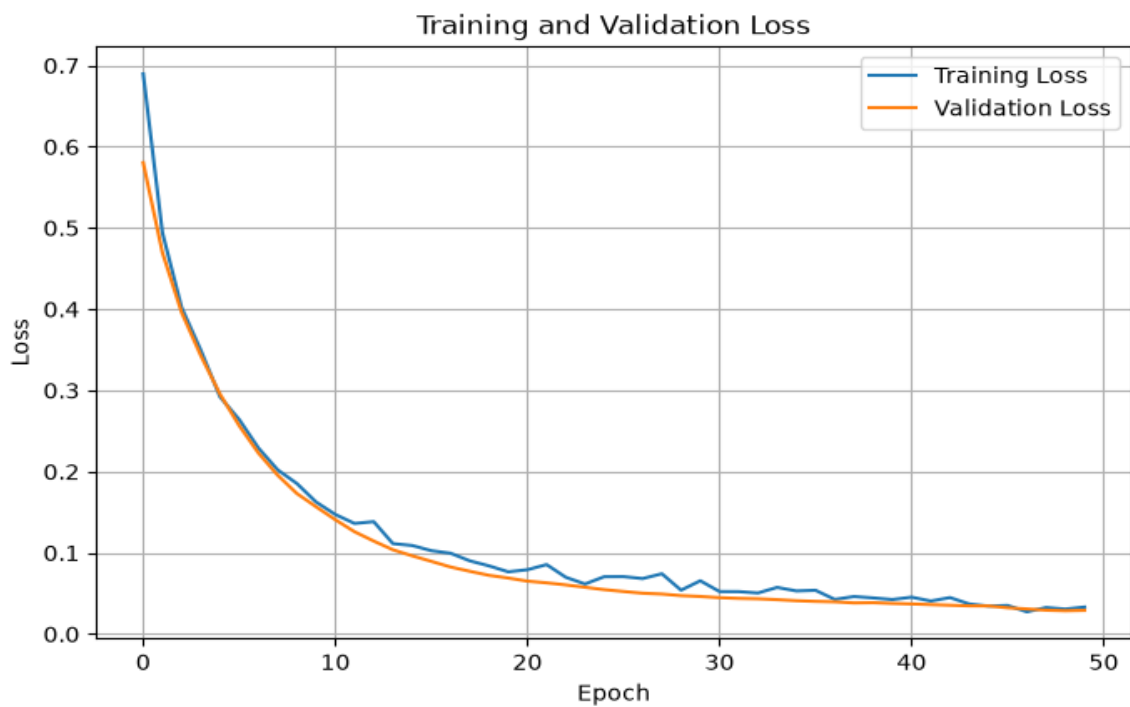


Figure 2. Training and Validation Loss

13. Model Evaluation

Metric	ANN Result
Test Loss	0.105609
Test Accuracy	94.74%

The ANN achieved 94.74% accuracy on the 114-sample test set. The recorded test loss was 0.105609.

14. Classification Report

Class	Precision	Recall	F1-score	Support
0	0.91	0.95	0.93	42
1	0.97	0.94	0.96	72
Accuracy			0.95	114
Macro avg	0.94	0.95	0.94	114
Weighted avg	0.95	0.95	0.95	114

15. Confusion Matrix

The confusion matrix shows 40 correctly predicted samples for class 0, 2 class-0 samples predicted as class 1, 4 class-1 samples predicted as class 0, and 68 correctly predicted class-1 samples. Therefore, the matrix contains 108 correct predictions out of 114 test samples.

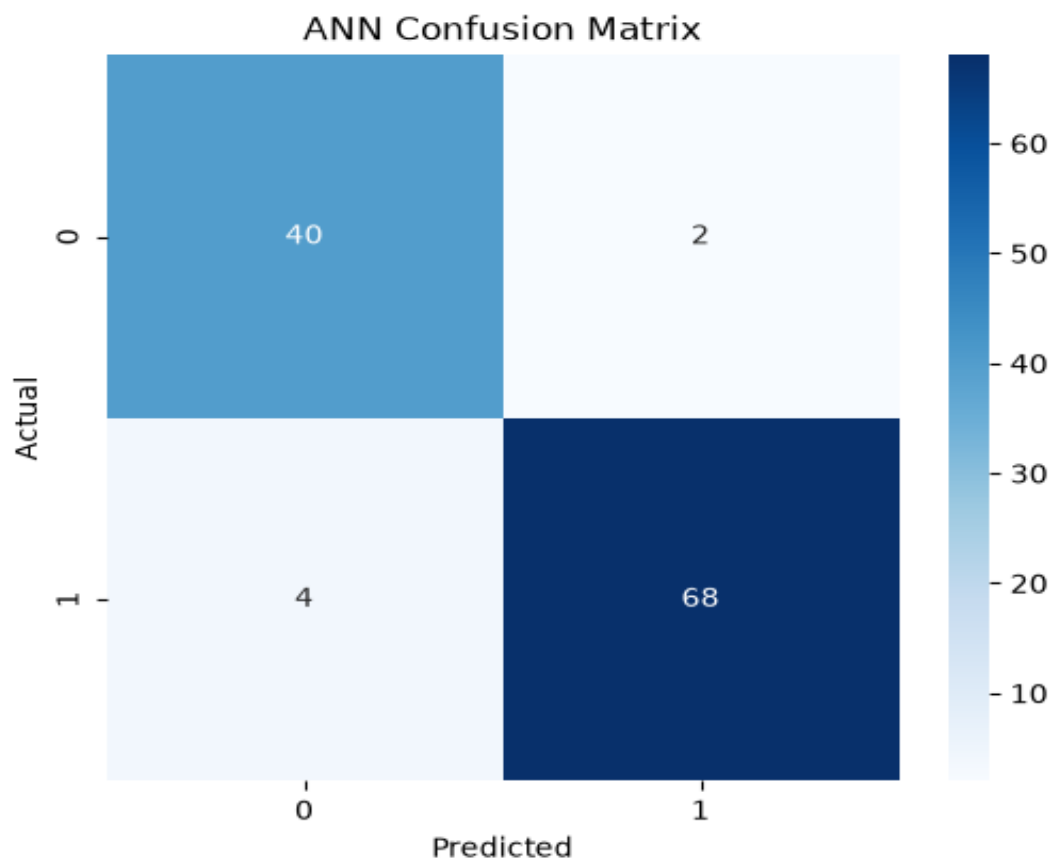


Figure 3. ANN Confusion Matrix

16. Overfitting and Regularization

Overfitting occurs when a model learns the training data too closely and performs less effectively on unseen data. The project uses a 20% Dropout layer as a regularization technique. The training and validation curves remain relatively close, which is a positive indication for generalization in the recorded training run.

17. ANN vs Traditional Machine Learning

Model	Accuracy
Logistic Regression	98.25%
Artificial Neural Network	94.74%

Logistic Regression outperformed the ANN on this recorded evaluation split. Its accuracy was 98.25%, compared with 94.74% for the ANN. The result demonstrates that model complexity does not guarantee better performance; model selection should be based on experimental evaluation.

18. Project Outputs and Saved Artifacts

Artifact	Purpose
task6_ann.ipynb	Complete interactive notebook
train_model.py	Standalone training/evaluation script
model.keras	Saved trained ANN model

Artifact	Purpose
scaler.pkl	Saved feature scaler
README.md	Project documentation
requirements.txt	Python dependencies
.gitignore	Git exclusions
results/	Accuracy, loss and confusion-matrix plots

19. Key Learnings

- Neural-network architecture and layer functions.
- ReLU and Sigmoid activation functions.
- Forward propagation and backpropagation.
- Gradient-based optimization using Adam.
- Feature scaling for neural-network training.
- Dropout regularization and validation monitoring.
- Classification evaluation using precision, recall, F1-score and confusion matrix.
- Comparison of ANN and traditional machine learning.

20. Conclusion

The Task 6 project successfully demonstrates the practical application of Deep Learning fundamentals through an Artificial Neural Network for breast cancer classification. The model used standardized input features, two ReLU hidden layers, Dropout regularization and a Sigmoid output layer. The ANN achieved **94.74% test accuracy** and **0.105609 test loss**.

The project also demonstrates the importance of model comparison. Logistic Regression achieved **98.25%** accuracy on the recorded evaluation split, exceeding the ANN's 94.74%. The overall workflow provides practical experience in preprocessing, neural-network construction, training, regularization, evaluation, visualization and model selection.

21. GitHub Submission

Recommended repository name: AIML-Task6-Deep-Learning-ANN

The GitHub repository should contain the notebook, Python script, README, requirements file, trained model, scaler and results folder. The local myenv/ virtual environment should remain excluded using .gitignore.

22. References

- Task 6 internship brief: Deep Learning Fundamentals & Neural Network Implementation.
- Scikit-learn Breast Cancer dataset.
- TensorFlow / Keras learning resources recommended for the project.

Data integrity note: Reported metrics and confusion-matrix values are taken from the project results provided for this report. The three uploaded project graphs are included directly in the report.